

A simulator for adaptive tutoring

Student models, knowledge tracing, question selection, and what we can and cannot measure with them

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1. What the simulator is for

A tutoring system has to choose the next question without being able to see what the student knows. To study that choice we need a student whose knowledge we *can* see, so that any claim about the tutor can be checked against the truth rather than against another estimate. Everything here is built around that one idea: the simulator owns a hidden state, the tutor never sees it, and every measurement compares the two.

The simulator has three replaceable parts, and keeping them separate is what makes the measurements interpretable.

Part	Sees the truth?	Question it answers
Student model	it <i>is</i> the truth	what does the learner actually know, and how does that change?
Knowledge tracing	no	what does the tutor believe, given only right and wrong answers?
Question selection	no	given that belief, what should be asked next?

Because they are separate, one can be varied with the others held fixed. That is the only way to attribute an effect: if the student and the tracer are fixed and only the scheduler changes, any difference is the scheduler's.

2. The curriculum

A curriculum is a directed acyclic graph of concepts. Each concept carries a tier (its depth), a list of prerequisites, and a bank of practice questions. Curricula are data, stored as JSON, so a new one is a new file rather than a code change.

The default graph is deliberately abstract and small: eight concepts named by tier, **A1**, **A2** at the top with no prerequisites, then **B1**, **B2**, **B3**, then **C1**, **C2**, and **D1** as the single capstone. A name states its own depth. The shape is a symmetric diamond, and two properties of it matter. The zone of proximal development starts at exactly two concepts, small enough to read at a glance; and **B2** is a prerequisite of both **C1** and **C2**, so a single weak concept visibly blocks two downstream. A 40-concept language curriculum is kept alongside it so that earlier results stay reproducible.

3. The student models

3.1 The plain BKT student

Each concept is a hidden coin: known, or not. Four parameters govern it.

	Meaning	Default
$p(L_0)$	probability the concept is known before any question	0.15
$p(T)$	probability an unknown concept becomes known at an attempt	0.20
$p(G)$	probability of a correct answer while it is unknown	0.25
$p(S)$	probability of a wrong answer while it is known	0.10

At each attempt on concept c , with $K_c \in \{0, 1\}$ the hidden state:

$$\Pr(\text{correct}) = \begin{cases} 1 - p(S), & K_c = 1, \\ p(G), & K_c = 0, \end{cases} \quad \text{then} \quad K_c \leftarrow 1 \quad \text{with probability } p(T) \text{ if } K_c = 0. \quad (1)$$

The student answers from the state it was in, and only then transitions. A learning event therefore shows up on the *next* question, not the one that caused it, which is why a student can be seen learning a concept on a question they got wrong. This ordering is not cosmetic: it is the ordering the tutor’s update assumes, and getting it backwards silently breaks calibration.

There is no forgetting, no question difficulty, and by default no effect of prerequisites on learning. That is the point. It is the simplest student for which the tutor’s arithmetic has a known right answer.

3.2 Two student models: S1 and S2

S1 Prerequisites do not affect learning. A concept is learned at $p(T) = 0.2$ whenever it is asked, whatever else the student knows.

S2 A concept is learned at $p(T) = 0.2$ once *all* of its prerequisites are truly known, and at $p(T) = 0.05$ otherwise.

S2 is the smallest change that makes the order of questions matter. Under S1 a question is worth the same whenever it is asked, so the order can buy little; under S2 a question asked too early is worth a quarter as much. The S1 effect is not exactly zero, and §7.4 reports its size and sign; the point is that it is small enough that a single run cannot resolve it.

3.3 The continuous student

The original student is also still available: mastery is a real number in $[0, 1]$, questions carry difficulty, prerequisites gate the learning rate continuously, and knowledge decays with elapsed time. It is the right model for studying forgetting and spaced practice, and the wrong one for checking the tutor’s arithmetic, because it has no binary truth to compare a probability against.

4. Knowledge tracing

The tutor keeps one number per concept: b_c , its belief that the student knows c . It sees only right and wrong answers. After an observation X it applies Bayes' rule and then the learning transition:

$$b^+ = \begin{cases} \frac{b(1-p(S))}{b(1-p(S)) + (1-b)p(G)}, & X = \text{correct}, \\ \frac{bp(S)}{bp(S) + (1-b)(1-p(G))}, & X = \text{wrong}, \end{cases} \quad b_c \leftarrow b^+ + (1-b^+)p(T). \quad (2)$$

Timing. The belief written by this update is the belief about the state the student is in *going into the next question*, because the transition has been applied. The student's hidden state after its own transition refers to the same instant. Any comparison of belief against truth must use that pairing; reading the state before the transition instead shifts everything by one step and makes a correct tracer look badly calibrated. This is the single easiest way to produce a convincing but wrong result, so it is stated here and enforced in the tests.

Because the tutor's parameters can be set independently of the student's, the tutor can be made deliberately wrong, which is what Section 7 exploits.

5. Question selection

The zone of proximal development is the set of concepts the tutor believes are worth teaching now:

$$\text{ZPD}(t) = \{ c : b_c < \theta \wedge b_p \geq \theta \ \forall p \in \text{prereq}(c) \}, \quad (3)$$

for a mastery threshold θ . Note that it is defined entirely on *beliefs*: the tutor has no access to the student's real state.

Two rules are compared.

Q2, `uniform_zpd` uniform over the ZPD, so a concept is not asked until its prerequisites are believed mastered. When nothing is prerequisite-ready, which in this graph means everything is believed mastered, it falls back to revisiting any concept.

Q1, `uniform_unmastered` uniform over every concept not yet believed mastered, ready or not.

Both are deliberately free of any bandit. The comparison is about prerequisite gating alone.

6. What is measured, and against what

Every metric reads the student's hidden state, never the tutor's belief.

Calibration. Bin every belief the tutor has ever held; within a bin, what fraction of the time was the student truly in the known state? A correct tracer lies on the diagonal.

Detection delay. We know the exact step at which each concept was learned. How many further questions *on that concept* pass before the tutor declares it mastered?

False alarm rate. Of all the declarations the tutor made, what share came before the student had actually learned the concept?

Concepts truly known. How many concepts the student really knows at the end. This is the only metric that can compare schedulers, because it owes nothing to what any tutor believes.

Every comparison is paired: the same seeds are used in every arm, so learner i starts from the same hidden state in every arm and differences are read within-learner rather than between groups. The pairing is on that shared starting state, not on every coin thereafter: once two schedulers ask different concepts the learners’ random draws diverge, so the pairing buys correlation rather than identity.

7. What the simulator has been used to establish

The tracer is correct. On the plain BKT student with a correctly specified tutor, pooled mean belief is 0.771 against a true known-fraction of 0.773 over 12,000 observations (100 learners, 120 questions each). At four times the learners, 48,000 pairs, all 8 bins holding at least 20 of the 48,000 observations sit within 0.029 of the diagonal. Sweeping each of the four parameters in turn over 19 settings leaves the pooled gap within 0.012 throughout. That last sweep keeps the tutor correctly specified in every cell, so it shows the estimator is unbiased across parameter settings; it is not by itself evidence that a *misspecified* tutor would be caught.

The two checks agree with each other. If beliefs are calibrated, the false-alarm rate at threshold θ must equal one minus the mean belief at the moment of declaration. They agree to within 2.6 percentage points at every one of the 7 thresholds, on 240 declarations per threshold which is inside sampling error at that many declarations. Two quantities computed by different routes landing this close is stronger evidence than either alone, though it is agreement within noise rather than identity.

The scheduler only matters if the student model says it can. Crossing S1 and S2 with Q1 and Q2, 400 learners, 40 questions, scored on concepts the student truly knows out of eight:

	Q1	Q2	Q2 – Q1	95% CI
S1	6.15	6.00	–0.15	± 0.14
S2	4.39	5.86	+1.47	± 0.18
interaction (S2 – S1)			+1.62	± 0.16

The interaction is far from zero: the scheduler is worth about one and a half concepts to S2 and nothing useful to S1. Because seed i is the same simulated person in all four cells, that interval is the paired one; the two per-student differences are correlated, so treating them as independent would overstate the uncertainty: $r = +0.49$, and the interval would be ± 0.23 rather than ± 0.16 if the two were treated as independent. The S1 figure is small and, with three contrasts in the table, marginal. It should be read as at most a slight cost, not as an established effect. A separate run at a longer horizon puts the S1 contrast at about one question in fifty-five, in the same direction; see §7.4.

What the restriction is actually worth. Q1 and Q2 differ in two ways at once, and separating them changes the story. Running a matched pair in which *both* arms gate on the student’s true state, so that only the prerequisite restriction differs:

	unrestricted	restricted	effect
S1	7.480	7.480	$+0.000 \pm 0.000$
S2	5.513	7.480	$+1.968 \pm 0.167$

Under S1 the restriction is *exactly* neutral: identical for all 400 learners, not merely close. So the small negative figure in the table above is not the cost of restricting to prerequisite-ready concepts. It is the cost of deciding readiness from a *belief that lags the truth*: Q2 withholds concepts the student is in fact already able to learn. The whiteboard prediction that the restriction would be invisible under S1 was right about the restriction; what it did not anticipate is the separate price of gating on an estimate.

Budget decides whether any of this is visible. At 120 questions the cells reach 7.99, 7.99, 7.68 and 7.90 of eight. The S1 effect is gone (-0.003 ± 0.015), but the S2 effect is still there and still positive ($+0.212 \pm 0.094$). So a longer budget shrinks the contrast towards the ceiling rather than abolishing it, and any comparison of question-selection rules has to report the budget it was measured at.

Misspecification degrades gracefully but not harmlessly. Sweeping the tutor’s assumed $p(G)$ against a fixed student, 200 learners, with intervals bootstrapped over learners: the pooled gap is $+0.002 \pm 0.006$ at the truth and -0.063 ± 0.008 when the tutor underestimates guessing by 0.20, while false alarms move from 4.6% to 17.1%. With a population whose parameters are spread uniformly by ± 0.15 and a single fixed tutor, the gap is -0.017 ± 0.014 and false alarms rise from 4.6% to 7.5%. A single tutor model stays usable across a varied population, at a measurable cost. Note the pooled gap is a signed average, so it is a bias diagnostic: opposite-signed bins can cancel, and per-bin behaviour is what Section 6’s calibration check looks at.

The same result, read in questions rather than concepts. Scoring on time to full mastery instead of on concepts known at a fixed budget gives the same verdict in a unit that is easier to read. Four hundred learners, three hundred questions, the median number of questions before a learner truly knows all eight concepts:

	Q1	Q2	mean diff	median diff	favours
S1	54	54	$+0.3 \pm 2.3$	+0.0	unresolved at this n
S2	78	55	-22.4 ± 4.3	-17.5	Q2

Every arm reaches all eight inside the budget, so nothing is censored and these are not finisher-only medians. The S2 contrast is censored and these are not finisher-only medians. A single run reports an interval, which says how well that sample pins the number down, not whether the sign would survive another sample. Re-running on independent blocks:

S1 averages $+0.37 \pm 0.57$ questions of cost to Q2 over 6 independent blocks, ranging from -0.93 to $+1.08$, but the blocks do not agree in sign, so the direction is still unsettled; S2 averages -23.08 ± 1.36 questions sooner under Q2 over 6 independent blocks, ranging from -25.61 to -20.66 , and every block agrees in sign, so the direction is settled.

So the S2 contrast is settled and the S1 contrast is not. Six blocks put S1 slightly on the costly side of zero, which agrees in direction with the concepts-known figure in §7.3, but the blocks disagree among themselves and the across-block interval covers zero. The honest statement is that the S1 effect is small enough that we cannot give it a sign, not that it is absent. An earlier draft read it off one block, found a negative number whose interval covered zero, and called it costless; that asserted a sign the data never supported, and in the direction the point estimate now contradicts. Note also that every arm asks the same fixed number of questions, so what differs is time to latent mastery, not effort spent.

What a single tutor should assume about a spread of students. If learners differ in one parameter and the tutor must use one value for everybody, which value, and how much would a

tutor that knew each learner’s own parameter gain over it? The second question is answerable only against an *oracle* arm handed each learner’s true parameters. No tutor can be built that way; the oracle is an upper bound on adapting, not a proposal. Six hundred learners per cell, one hundred and twenty questions:

vary	spread	best fixed	its Brier	oracle	buys	95% CI	resolved
p-S	± 0.05	0.10	0.0835	0.0834	+0.0002	± 0.0005	no
p-S	± 0.10	0.10	0.0836	0.0834	+0.0002	± 0.0013	no
p-S	± 0.15	0.10	0.0854	0.0835	+0.0019	± 0.0013	yes
p-G	± 0.05	0.25	0.0826	0.0832	−0.0006	± 0.0016	no
p-G	± 0.10	0.25	0.0824	0.0835	−0.0011	± 0.0018	no
p-G	± 0.15	0.25	0.0829	0.0828	+0.0001	± 0.0020	no

The best single fixed tutor is the one sitting at the true centre, at every spread and for both parameters. Adapting to $p(S)$ buys a real but small improvement, and only once learners are widely spread: about two percent of the score at spread ± 0.15 , with nothing resolvable at ± 0.05 or ± 0.10 . Adapting to $p(G)$ buys nothing detectable at any spread tried, and is nominally negative at two of three.

The score is the Brier score, and the choice matters more than it looks. The pooled signed calibration gap used elsewhere in this document *cancels*: a tutor overconfident about half its learners and underconfident about the other half scores zero, and under a symmetric spread that is exactly the error adapting would remove, so scoring this study on it would make it blind to its own question. Mean absolute error does not cancel but is improper, minimised by the median, so while most concepts are still unknown it rewards a tutor that simply stays pessimistic: with every learner at the truth it prefers $p(S) = 0.02$ to the true 0.10. Brier is proper and does recover the true parameter at this horizon. Both rejected scores are kept in the CSV as diagnostics. An earlier draft scored this study on the signed gap at sixty learners and concluded that adapting buys nothing; that conclusion was an artifact of the score and the sample size.

A caution about reading parameter sensitivity. A concept is declared mastered when the belief crosses θ , which takes an integer number k of consecutive correct answers. Over the swept grids k runs 2, 2, 3, 3, 4 for $p(G)$, 3, 3, 3, 3, 3 for $p(S)$ and 3, 3, 3, 3, 2 for $p(T)$. $p(S)$ never crosses a regime boundary at all, so its apparent insensitivity for premature declarations is partly a property of the grid rather than of the parameter. Sensitivity comparisons across parameters here are closer to counting threshold crossings than to derivatives, and should be expected to move with θ or with $p(L_0)$.

8. Limitations

The plain BKT student has no forgetting, so nothing here speaks to retention or spaced practice; the continuous student exists for that and is used in none of the checks above. All results are on one curriculum of eight concepts. The 40-concept graph ships alongside it and is loadable, but no result in this document has been rerun on it, so nothing here should be read as established beyond the small graph. Every population in this document derives from a single base seed, so the intervals describe sampling within one seed block rather than across independent replications. Per-bin calibration intervals are nominal: successive observations on one concept are dependent, because the belief after one attempt is the belief before the next, so the pooled comparison is the robust one. The threshold $\theta = 0.9$ is shared between the tracer and the scheduler, which means it sets both when a concept is declared and when its dependants become available, and it is not swept.

The tutor's $p(T)$ against S2 is held constant even though the student's varies with its prerequisites, so the tutor is mildly misspecified there by construction; whether it should instead gate $p(T)$ on its own belief about the prerequisites is open. And every number is a statement about this model, not about people: no result here has been checked against a real learner.

Reproducing. `python3 make_sanity.py` regenerates every figure and table quoted above into `sanity_outputs/`, and `python3 make_tables.py` turns those into every table in this document, so no number here is typed by hand; `python3 make_tables.py --check` fails if the two ever disagree. `python3 test_sanity.py` and `python3 test_simulator.py` run the two suites, including the exactness checks that pin the update ordering (with $p(G) = p(S) = 0$ a correct answer must come from a known concept; with $p(G) = p(S) = 0$, $p(L_0) = 0$ and $p(T) = 1$ nothing is known before its first question and one attempt makes it known with certainty, so the tutor's belief equals the truth at every step). Seeds are fixed throughout.